Geometrical Instruments in Practice: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IOAA questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

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Solution 1 — Separation from plate scale. Return →

The separation is $8 \times 1 = 8''$. **Check:** plate scale is a direct conversion from chart length to angle.

Solution 2 — Position angle. Return →

Due east is 90°; due north is 0°. **Check:** position angle is measured from north through east.

Solution 3 — Components. Return →

$\Delta\alpha \cos\delta = 8 \sin 60^\circ = 6.93''$ and $\Delta\delta = 8 \cos 60^\circ = 4.00''$. **Check:** the east component is larger because the angle is closer to east than north.

Solution 4 — Reading uncertainty. Return →

At 2″/mm, the uncertainty is $0.5 \times 2 = 1''$. At 0.5″/mm, it is $0.5 \times 0.5 = 0.25''$. **Check:** a smaller plate scale k (arcsec/mm) reduces the angular effect of the same ruler error.

Solution 5 — Calibrating scale. Return →

The plate scale is $10/5 = 2''$/mm. The target separation is $12 \times 2 = 24''$. **Check:** calibration transfers a known angular scale to the chart.

Solution 6 — Mirrored chart. Return →

An east–west mirror changes P to $360^\circ - P$, so 60° becomes 300°. **Check:** the north component is unchanged, but the east component reverses sign.

Solution 7 — Position-angle uncertainty. Return →

$\sigma_P \approx 0.1/5 = 0.02$ rad, or 1.15°. **Check:** closer pairs have larger angular uncertainty for the same positional error.

Solution 8 — Resolution limit. Return →

No. A 0.8″ pair is below the 1.5″ practical resolution, so it cannot be reliably measured as two components. Report an upper limit on separation or note that the pair is unresolved. **Check:** a false measurement is worse than an honest limit.

Solution 9 — Epoch and motion. Return →

Under the constant-rate approximation, the change is $2 \times 10 = 20^\circ$. The epoch must be stated because orbital motion changes the position angle over time, and the rate can vary around an eccentric orbit. **Check:** a measurement without an epoch cannot be compared with other observations.

Solution 10 — Design a measurement. Return →

Align the protractor’s zero with chart north, measure the centre-to-centre length with the ruler, convert using the plate scale, and read the position angle. Repeat the measurements, rotate the chart to check alignment, and use a fine-scale chart or magnifier. Record the ruler and protractor resolution and propagate them into the final uncertainty. **Check:** repeated independent readings reveal alignment and reading errors.